

Predicting Credit Card Fraud

Complete Machine Learning Assignment Report

FinGuard Analytics

1. Problem Overview

FinGuard Analytics is experiencing increasing financial losses due to fraudulent credit card activity. This project develops a complete machine learning pipeline to identify fraudulent transactions with high reliability while maintaining efficiency for real-time deployment.

This is a binary classification problem:

- 0 = Legitimate transaction
- 1 = Fraudulent transaction

The primary challenge is extreme class imbalance — fraud represents only 0.17% of all transactions, making standard approaches unreliable without special handling.

2. Dataset Overview

Property	Detail
Total Transactions	284,807
Total Features	31
Missing Values	None
Legitimate Transactions	284,315 (99.83%)
Fraudulent Transactions	492 (0.17%)

2.1 Key Data Insights

- Average legitimate transaction: \$88.29
- Average fraudulent transaction: \$122.21
- Largest fraud transaction: \$2,125.87
- V1-V28 features are anonymized by the bank using PCA for privacy
- Amount alone is a weak fraud signal despite the \$33.92 average difference

3. Data Preprocessing

3.1 Feature Scaling

StandardScaler was applied to the Amount and Time columns to bring them into the same range as V1-V28. Without scaling, large dollar values in Amount would incorrectly dominate model decisions.

3.2 Train-Test Split

- Training set: 227,845 transactions (80%)
- Testing set: 56,962 transactions (20%)

3.3 SMOTE — Handling Class Imbalance

SMOTE (Synthetic Minority Over-sampling Technique) was applied to generate realistic synthetic fraud examples in the training data.

	Before SMOTE	After SMOTE
Legitimate	227,451	227,451
Fraud	394	227,451

4. Model Development

4.1 Evaluation Metrics

Given the extreme class imbalance, these metrics were prioritised over accuracy:

- Precision — Of all transactions flagged as fraud, how many were actually fraud?
- Recall — Of all actual fraud transactions, how many did the model catch?
- F1 Score — Harmonic mean of Precision and Recall
- AUC — How well the model ranks transactions by fraud risk

4.2 Model 1 — Logistic Regression (Baseline)

Metric	Legitimate	Fraud
Precision	1.00	0.06
Recall	0.97	0.93
F1 Score	0.99	0.11
AUC	-	0.979

Despite high recall (93%), precision of just 6% makes this model unsuitable. For every 100 fraud flags, 94 are innocent customers wrongly accused.

4.3 Model 2 — Random Forest

Metric	Legitimate	Fraud
Precision	1.00	0.93
Recall	1.00	0.82
F1 Score	1.00	0.87
AUC	-	0.942

Random Forest delivered dramatically superior results with 93% precision and 0.87 F1 Score — the best deployment model.

4.4 Model 3 — XGBoost (Tuned)

Metric	Legitimate	Fraud
Precision	1.00	0.20
Recall	0.99	0.90
F1 Score	1.00	0.32
AUC	-	0.988

XGBoost achieved the highest AUC score of 0.988, making it the best model for ranking transactions by risk level. However its lower F1 makes Random Forest preferable for direct fraud blocking decisions.

4.5 Full Model Comparison

Model	Precision	Recall	F1 Score	AUC
Logistic Regression	6%	93%	0.11	0.979
Random Forest	93%	82%	0.87	0.942
XGBoost Tuned	20%	90%	0.32	0.988

5. Cross Validation

Five-fold cross validation was performed to test model reliability across different data splits.

Model	Average F1	Consistency
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Logistic Regression	0.948	+/- 0.001
Random Forest	1.000	+/- 0.000
XGBoost Tuned	0.990	+/- 0.001

Random Forest achieved a perfect cross-validation F1 of 1.000 with zero variance — meaning it performed perfectly across all 5 test splits. The slight drop from 1.000 to 0.87 on unseen test data indicates minor overfitting, but the model remains excellent in practice.

6. Principal Component Analysis (PCA)

6.1 PCA Configuration

PCA was configured to retain 95% of data variance, automatically reducing the 30 features to fewer principal components.

6.2 Performance Impact

Metric	Without PCA	With PCA
Fraud Precision	93%	63%
Fraud Recall	82%	85%
Fraud F1 Score	0.87	0.72

PCA reduced precision from 93% to 63% — an unacceptable drop for fraud detection. While PCA offers faster training, the cost to customer experience is too high. The full 30-feature model is recommended for deployment.

7. Feature Importance

Rank	Feature	Importance	Signal
1	V14	41.3%	Dominant
2	V17	13.7%	Strong
3	V10	10.0%	Strong
4	V12	7.8%	Significant
5	V2	5.0%	Moderate
6	V4	3.5%	Moderate
9	Amount	1.2%	Weak

- V14 alone drives 41.3% of all fraud detection decisions
- Top 4 features combined account for 73% of total detection power
- Amount is a surprisingly weak signal at only 1.2% importance
- Bottom features V22, V24, V27 contribute less than 0.5% each

8. Business Impact Analysis

Based on 56,962 test transactions with an average fraud value of \$122.21:

Metric	Random Forest	XGBoost Tuned
Fraud Cases Caught	80	88
Fraud Cases Missed	18	10
Innocent Customers Wrongly Flagged	6	358
Money Saved	\$9,776.80	\$10,754.48
Money Lost to Missed Fraud	\$2,199.78	\$1,222.10

While XGBoost saves \$977.68 more per cycle, it wrongly flags 358 innocent customers versus only 6 for Random Forest. Considering customer trust, operational costs of handling false positives, and potential regulatory concerns, Random Forest is the superior business choice despite the slightly lower financial recovery.

9. Real-Time Transaction Prediction

The system was tested on individual transactions to simulate real-time deployment:

Test	Actual	LR Prediction	RF Prediction	XGB Prediction
Legit Transaction	LEGITIMATE	LEGIT (2.6%)	LEGIT (0.0%)	LEGIT (1.1%)
Fraud Transaction	FRAUD	FRAUD (100%)	FRAUD (90%)	FRAUD (100%)

All three models correctly identified both the legitimate and fraudulent transaction. Random Forest's 0.0% fraud probability on the legitimate transaction demonstrates exceptional confidence in clearing innocent customers.

10. Discussion of Trade-offs

Trade-off	Finding
Performance vs Interpretability	Random Forest outperforms Logistic Regression significantly but is harder to explain. Logistic Regression is simpler but performs poorly.
Performance vs Speed (PCA)	Removing PCA preserves precision (93% vs 63%). Speed gains from PCA are not worth the performance cost.
Recall vs Precision	Higher recall catches more fraud but flags innocent customers. Random Forest provides the best balance.
AUC vs F1	XGBoost leads on AUC (0.988) making it best for risk scoring. Random Forest leads on F1 (0.87) making it best for blocking decisions.
Money vs Customer Experience	XGBoost saves \$977 more but wrongly flags 358 innocent customers. Random Forest is the better business choice.

11. Conclusions & Recommendations

11.1 Final Recommendations

- Deploy Random Forest as the primary fraud blocking model — best F1, best precision, minimal false positives
- Use XGBoost as a risk scoring layer — highest AUC, best at ranking transactions by suspicion level
- Do not apply PCA in production — precision drop from 93% to 63% is unacceptable
- Monitor V14 closely — its 41.3% importance makes it the single most critical fraud signal
- Never use accuracy alone as a metric for imbalanced fraud data

11.2 Future Improvements

- Investigate the real-world meaning of V14 with domain experts
- Experiment with Neural Networks for fraud detection
- Build a REST API for real-time transaction scoring
- Implement cost-sensitive learning to directly optimise financial outcomes
- Remove low-importance features V22, V24, V27 to simplify the model
- Build a monitoring dashboard to track fraud pattern changes over time